

Automatic Grid-Following $\leftrightarrow$ Grid-Forming Mode Switching of One Synchronous Generator and Four IBRs on the IEEE 14-Bus System via AGSI++

Small-signal stability sweep, fault switching-outcome map, and time-domain
trip/reclose switching (project-owned MATLAB, no external solver)

24 July 2026

Abstract

This standalone report consolidates the IEEE 14-bus study of automatic grid-following (GFL) $\leftrightarrow$ grid-forming (GFM) mode switching driven by the Adaptive Grid-Support Index (AGSI++). The test system has **one synchronous generator** at bus 1 (Padiyar model-1.1, *manual field, no AVR*, 4-state) sharing a *meshed single-area* network with **four GFL IBRs** at buses 2, 3, 6 and 8 (reduced-6 SwitchableIbr6). The device models are identical to those of the companion Padiyar two-area switching study [1]. The composite small-signal model has **28 states** and **zero right-half-plane eigenvalues** ($\max \text{Re}(\lambda) \approx 0$); the least-damped mode is $f = 0.345$ Hz with $\zeta = +0.267$, so the operating point is *small-signal stable* and satisfies the no-AVR / no-PSS, 1-SG + 4-IBR constraint. The report brings together (A) an SSSA stability sweep over inertia and internal-reactance (coupling) scaling, (B) a fault switching-outcome map over fault severity and location, and (C) the power-flow operating point, the SSSA verdict, the time-domain SG trip/reclose switching cycle, and the AGSI++ weight-tuning provenance. All production dynamics are project-owned base MATLAB (audited lu/eig primitives only); external programs are never numerical acceptance targets.

1 Scope and link to the companion report

This report is standalone but reuses, as the single source of truth, the reduced-6 IBR models (GFL, GFM), the SwitchableIbr6 bumpless device, the Padiyar model-1.1 synchronous machine, and the AGSI++ supervisor documented and derived in full in the companion report `report_two_ibr_switch_en` [1, 2, 3, 4]. Here those models are applied to the meshed IEEE 14-bus system, and three connected artifacts are added: a small-signal stability sweep (§5), a fault switching-outcome map (§7), and the time-domain trip/reclose switching cycle (§6). Source discipline: the AGSI++ structure follows the EECON49-P4 strategy [1] as a design guideline (PROJECT_DERIVED_SOURCE_GUIDED); the diagnostic time-domain driver and the two sweeps are ASSUMED_DIAGNOSTIC; nothing was tuned to bias any result.

2 Models and the AGSI++ switching index

Switching index. The AGSI++ merges six normalised per-IBR sub-indices into one dimensionless index, evaluated every step:

$$\begin{aligned} J_V &= \frac{|V - V_{ref}|}{0.10}, & J_f &= \frac{|f - 60|}{0.50}, & J_R &= \frac{|df/dt|}{1.0}, \\ J_P &= \frac{|P_{ref} - P|}{0.20}, & J_{SCR} &= \max\left(0, \frac{3}{SCR} - 1\right), & J_{lock} &= \frac{|v_q|}{0.10}, \end{aligned} \quad (1)$$

$$AGSI = w_V J_V + w_f J_f + w_R J_R + w_P J_P + w_{SCR} J_{SCR} + w_{lock} J_{lock}, \quad (2)$$

with weights (sum = 1) $[w_V, w_f, w_R, w_P, w_{SCR}, w_{lock}] = [0.25, 0.25, 0.15, 0.10, 0.15, 0.10]$. Switching uses hysteresis and dwell: GFL→GFM when $AGSI \geq \Gamma_{on} = 0.65$ (held $\geq T_{d,on}$); GFM→GFL when $AGSI < \Gamma_{off} = 0.35$ (held $\geq T_{d,off}$). AGSI++ low-pass filters the RoCoF term ($\tau_R = 0.05$ s) to reject the one-sample RoCoF spike at a topology change, and adds the weak-grid strength term J_{SCR} and the PLL loss-of-lock proxy $J_{lock} = |v_q|/0.10$.

Reduced-6 IBR models. **GFL** state order $x = [i_d, i_q, \delta_{PLL}, \xi_{PLL}, \xi_P, \xi_Q]$ (L-filter dq currents, SRF-PLL angle and integrator, and P/Q outer-loop integrators). **GFM** state order $x = [i_d, i_q, \omega, \delta, E, \xi_V]$ (currents, virtual-rotor speed and angle, internal voltage magnitude, and d-axis voltage integrator), with cross-paired terminals $v_{gd} = E + X i_q$, $v_{gq} = -X i_d$. Each IBR is one `SwitchableIbr6` device wrapping both branches; the active state is always 6-dimensional and a switch is *bumpless* — the target branch is re-initialised to the instantaneous terminal (V, P, Q) so the injected current is continuous,

$$I^+ = \overline{\left(\frac{s}{V}\right)} = I^-. \quad (3)$$

Synchronous generator (Padiyar model-1.1, manual, 4-state). State order $[\delta, \omega, E'_q, E'_d]$:

$$\dot{\delta} = \omega_B(\omega - 1), \quad \dot{\omega} = \frac{P_m - (\omega - 1)/R - T_e - D(\omega - 1)}{2H}, \quad (4)$$

with $E_{fd} = E_{fd0}$ constant (no AVR), $\dot{E}'_q = [E_{fd} - E'_q - (X_d - X'_d)I_d]/T'_{d0}$ and $\dot{E}'_d = [-E'_d + (X_q - X'_q)I_q]/T'_{q0}$. The manual (no-AVR, no-PSS) machine keeps the composite free of any excitation-control state, as required by the study constraint. Full derivations, the stator solution, and the network DAE are in the companion report.

3 Test system

Figure 1 is the single-line diagram of the study system: the IEEE 14-bus network with the synchronous generator **SG**₁ at bus 1 acting as slack (angle reference) and four inverter-based resources **IBR**₁–**IBR**₄ at buses 2, 3, 6 and 8, each backed by an energy source and/or storage. In steady state every IBR is dispatched as a PQ (grid-following) unit; when the AGSI++ index commands a switch, the affected unit becomes a voltage-forming (PV / grid-forming) source. The remaining buses carry the standard IEEE 14-bus loads, and the transformer taps of branches 4–7, 4–9 and 5–6 are represented exactly by the tap-aware admittance matrix taken from the

in-house power flow. This is the same one-line arrangement used in the EECON49-P4 study [1], which is why the device mapping (1 SG + 4 IBR at the original generator buses) is kept unchanged here.

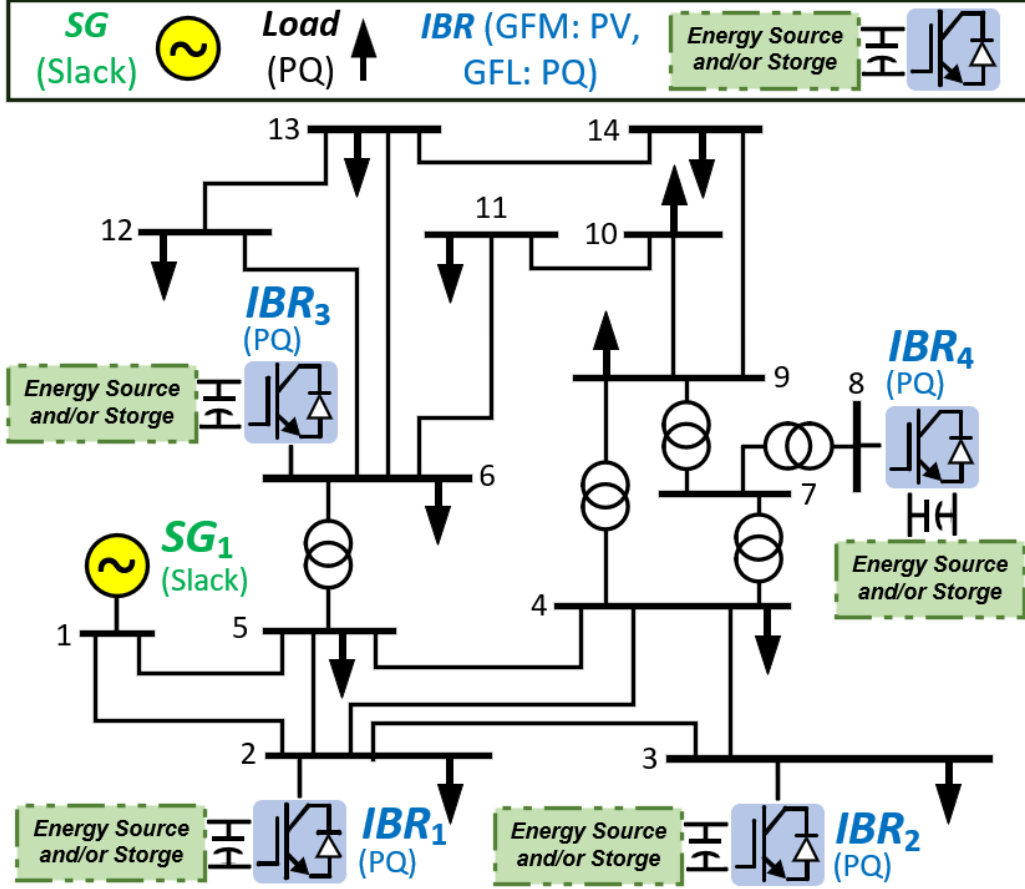


Figure 1: Single-line diagram of the IEEE 14-bus test system used in this study: SG₁ (slack) at bus 1 and the four switchable IBRs at buses 2, 3, 6 and 8 (GFM: PV, GFL: PQ), each with an energy source and/or storage; arrows are loads (PQ). Reproduced from the EECON49-P4 study system [1].

4 Power-flow operating point

The base case is solved by the in-house Newton–Raphson power flow. The synchronous generator at bus 1 carries the bulk dispatch, $P_{g1} = 2.324 \text{ pu} = 232.4 \text{ MW}$ (slack / angle reference). The IBR at bus 2 dispatches $P_g = 0.40 \text{ pu}$; the IBRs at buses 3, 6 and 8 are on voltage/reactive support duty with scheduled $P_g = 0$. Load is distributed over the remaining IEEE 14-bus buses. This operating point is the initial condition for both the small-signal linearisation (§5) and every time-domain run (§6); the pre-event active-power panel (Fig. 6) shows the SG holding $\approx 2.32 \text{ pu}$ and IBR1 holding $\approx 0.40 \text{ pu}$, confirming the dispatch.

5 Small-signal stability and the stability sweep

Linearising the composite 28-state model about the base case gives **zero right-half-plane eigenvalues**: $\max \text{Re}(\lambda) \approx 0$ (only the ≈ 0 angle-reference gauge mode, at the 10^{-8} level), and the least-damped oscillatory mode is $f = 0.345$ Hz with $\zeta = +0.267$. The meshed single-area IEEE 14-bus operating point is therefore *small-signal stable*.

To test how robust this verdict is, two one-parameter sweeps re-derive the equilibrium consistently and re-linearise at each point (Fig. 2, Table 1): an *inertia* sweep on H_x (scaling SG H and damping) and a *coupling* sweep on k_x (scaling SG internal reactances; larger = weaker coupling). The inertia sweep is essentially flat and stable across $H_x \in [0.02, 2]$ ($\max \text{Re} \sim 10^{-8}$, $\zeta \approx 0.267$, mode near 0.345 Hz). The coupling sweep stays stable across $k_x \in [1, 20]$: damping erodes *monotonically* ($\zeta : 0.267 \rightarrow 0.150$ at $20\times$ internal reactance) but never crosses $\zeta = 0$, and $\max \text{Re}$ stays at the 10^{-8} gauge level. **No instability boundary is found on either knob.**

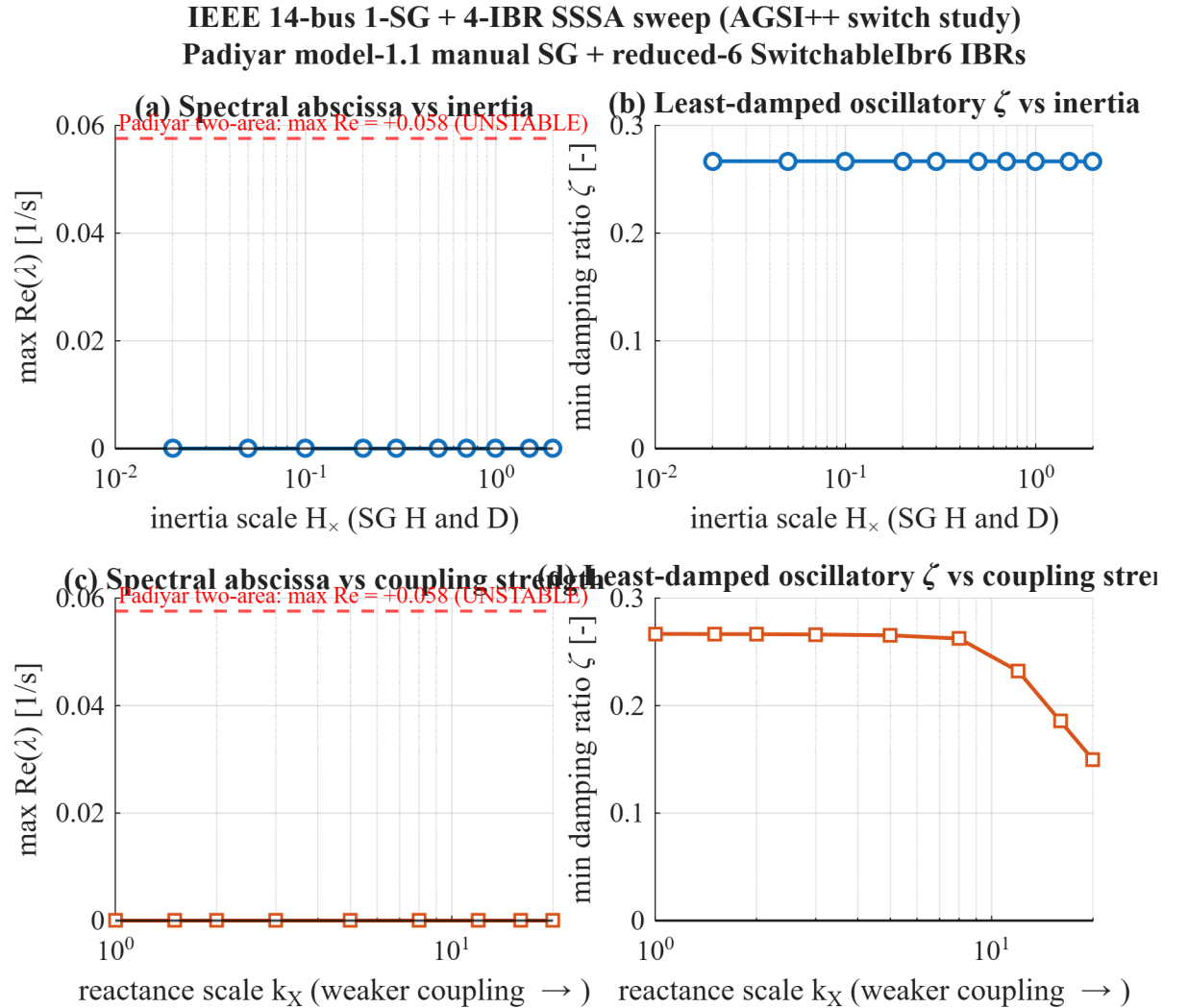


Figure 2: SSSA stability sweep of the IEEE 14-bus 1-SG + 4-IBR system: spectral abscissa $\max \text{Re}(\lambda)$ and least-damped ζ versus inertia scaling (top, H_x) and coupling/internal-reactance scaling (bottom, k_x). The IEEE 14-bus system stays stable throughout ($\max \text{Re} \approx 0$); the red dashed line marks the Padiyar two-area contrast at $\max \text{Re} = +0.058 \text{ s}^{-1}$ (UNSTABLE, a different weak-tie topology).

Table 1: SSSA stability sweep (inertia and coupling knobs) with the Padiyar two-area contrast. Generated entirely by `scripts/reporting/ieee14_sssa_sweep.m`.

Sweep	Param	max Re(λ) (1/s)	$\zeta_{\min}$ (-)	f (Hz)	Verdict (n_{unstable})
<i>IEEE14 inertia sweep ($H_{\times}$ scales SG H and D; $k_X = 1$)</i>					
$H_{\times} = 2.00$	2	$+3.222 \times 10^{-8}$	0.2665	0.3448	stable (0)
$H_{\times} = 1.50$	1.5	$+3.735 \times 10^{-8}$	0.2666	0.3448	stable (0)
$H_{\times} = 1.00$	1	$+4.448 \times 10^{-8}$	0.2667	0.3449	stable (0)
$H_{\times} = 0.70$	0.7	$+5.021 \times 10^{-8}$	0.2667	0.3449	stable (0)
$H_{\times} = 0.50$	0.5	$+5.492 \times 10^{-8}$	0.2667	0.3449	stable (0)
$H_{\times} = 0.30$	0.3	$+6.065 \times 10^{-8}$	0.2668	0.3449	stable (0)
$H_{\times} = 0.20$	0.2	$+6.396 \times 10^{-8}$	0.2668	0.3450	stable (0)
$H_{\times} = 0.10$	0.1	$+6.766 \times 10^{-8}$	0.2668	0.3450	stable (0)
$H_{\times} = 0.05$	0.05	$+6.971 \times 10^{-8}$	0.2668	0.3450	stable (0)
$H_{\times} = 0.02$	0.02	$+7.095 \times 10^{-8}$	0.2668	0.3450	stable (0)
<i>IEEE14 grid-coupling sweep (k_X scales X_d, X'_d, X_q, X'_q; $H_{\times} = 1$)</i>					
$k_X = 1.00$	1	$+4.448 \times 10^{-8}$	0.2667	0.3449	stable (0)
$k_X = 1.50$	1.5	$+5.836 \times 10^{-8}$	0.2666	0.3448	stable (0)
$k_X = 2.00$	2	$+6.357 \times 10^{-8}$	0.2664	0.3447	stable (0)
$k_X = 3.00$	3	$+6.290 \times 10^{-8}$	0.2662	0.3445	stable (0)
$k_X = 5.00$	5	$+5.070 \times 10^{-8}$	0.2654	0.3440	stable (0)
$k_X = 8.00$	8	$+3.453 \times 10^{-8}$	0.2624	0.3425	stable (0)
$k_X = 12.00$	12	$+2.004 \times 10^{-8}$	0.2320	0.3325	stable (0)
$k_X = 16.00$	16	$+7.427 \times 10^{-9}$	0.1855	0.3193	stable (0)
$k_X = 20.00$	20	$+1.179 \times 10^{-8}$	0.1497	0.3117	stable (0)
<i>Contrast: Padiyar two-area 1-SG+3-IBR (weak-tie topology)</i>					
Padiyar	—	$+5.758 \times 10^{-2}$	0.1224	0.6766	UNSTABLE (1)

Interpretation (topology, not inertia or coupling). The contrast with the Padiyar two-area system (1 SG + 3 IBR, weak inter-area tie) is decisive: that operating point has one right-half-plane eigenvalue, $\max \text{Re} = +0.058 \text{ s}^{-1}$ ($\zeta = +0.122$, $f = 0.677 \text{ Hz}$), i.e. *small-signal unstable*, whereas the meshed IEEE 14-bus stays stable even at $20\times$ internal reactance or $0.02\times$ inertia. The instability of the two-area case is therefore **topological** — it comes from the weak inter-area tie, not from low inertia or coupling strength, which a meshed grid tolerates.

6 Time-domain switching: SG trip and reclose

The demonstration run applies, over a 50 s window: a temporary three-phase fault $Z_f = j0.1 \text{ pu}$ at bus 4 during 1.0–1.15 s, an **SG trip at 6 s**, and a synchronised **SG reclose at 9 s**. The two disturbances are deliberately *well separated*: the fault is cleared 4.85 s before the breaker opens,

so the post-fault transient has fully decayed and the IBR response to the trip is not contaminated by the fault. (This differs from the shorter 6 s objective run used for weight tuning in §8, which trips at 1 s and recloses at 4 s.) Figures 3–6 show the AGSI++ per IBR, the bus voltages, the frequencies, and the active-power injections. The ordered timeline is:

1. $[0, 1)$ s — SG is the slack; all four IBRs run as GFL at their scheduled dispatch.
2. 1.0–1.15 s — temporary fault $Z_f = j0.1$ pu; the IBRs *ride through* it as GFL (AGSI spikes but the fault clears and no IBR reaches the dwell-confirmed Γ_{on} crossing).
3. 1.15–6 s — fault-free recovery; the trajectory settles back to the SG-slack operating point before the next event, so the trip is a clean, independent disturbance.
4. $t = 6$ s — **SG trips**. All four IBRs cross Γ_{on} (AGSI ≈ 1.88 –2.07) and **form GFM** after the $T_{d,on} = 0.5$ s dwell, i.e. at $t = 6.5$ s, so the island survives (network minimum voltage ≈ 0.49 –0.53 pu on the formed island; $\min |V| = 0.453$ pu at the trip transient).
5. $t = 9$ s — **synchronised reclose**; the angle reference is handed back to the SG and all four IBRs revert to their scheduled GFL dispatch. The reclose transient briefly re-triggers forming at $t = 9.5$ s, and AGSI++ peaks bounded at ≈ 3.5 .
6. $t \rightarrow 50$ s — the response *stays bounded*: voltages recover ($V_{end} \approx 0.941$ pu at the weakest bus, higher elsewhere), the frequency excursion peaks near 61.4 Hz and then settles at ≈ 59.6 Hz (no governor restores 60 Hz exactly), the SG re-takes ≈ 2.53 pu, and three of the four IBRs revert to GFL as their index falls below Γ_{off} at $t \approx 22.3$ –23.1 s (IBR2 at bus 3 remains GFM, supporting the weakest node). AGSI++ stays bounded (≈ 3.5 peak) with no chatter.

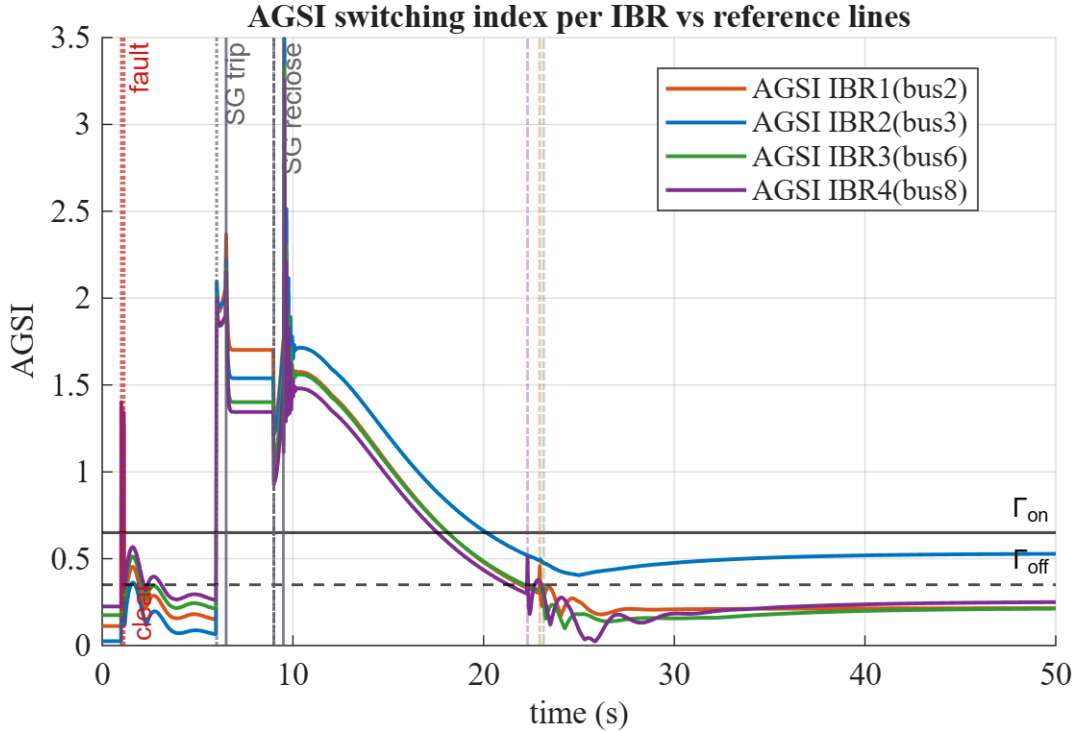


Figure 3: IEEE 14-bus: AGSI++ per IBR versus $\Gamma_{on} = 0.65 / \Gamma_{off} = 0.35$ (fault $Z_f = j0.1$ pu at 1.0–1.15 s, SG trip 6 s, reclose 9 s; 50 s window).

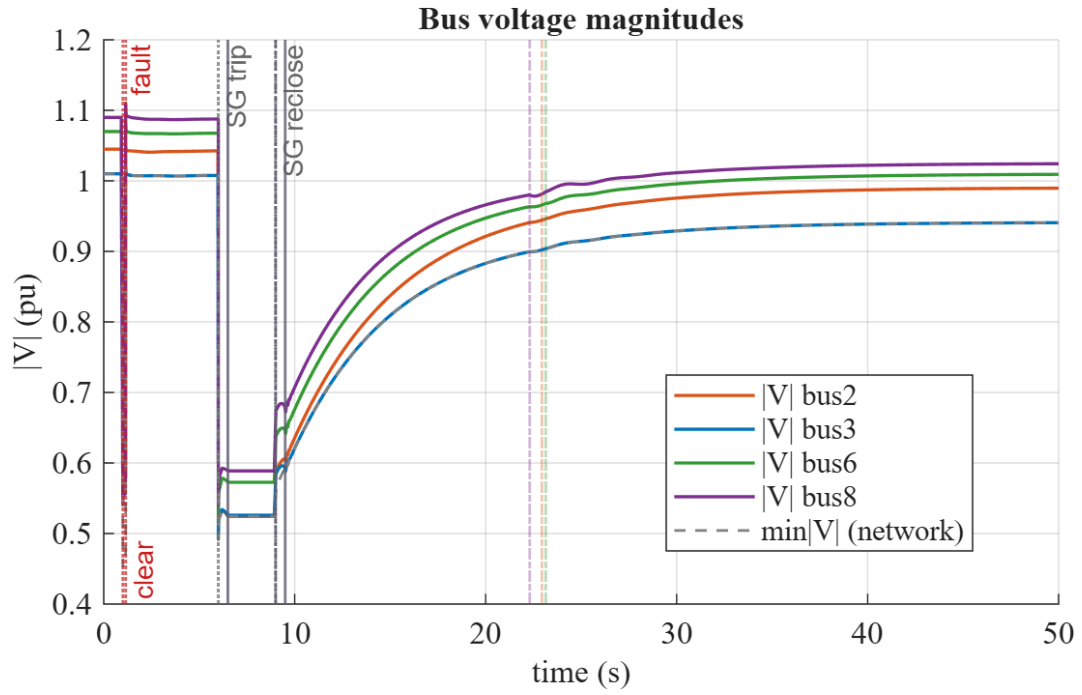


Figure 4: IEEE 14-bus: bus-voltage magnitudes through the fault / trip / reclose cycle; the network minimum recovers to ≈ 0.94 pu.

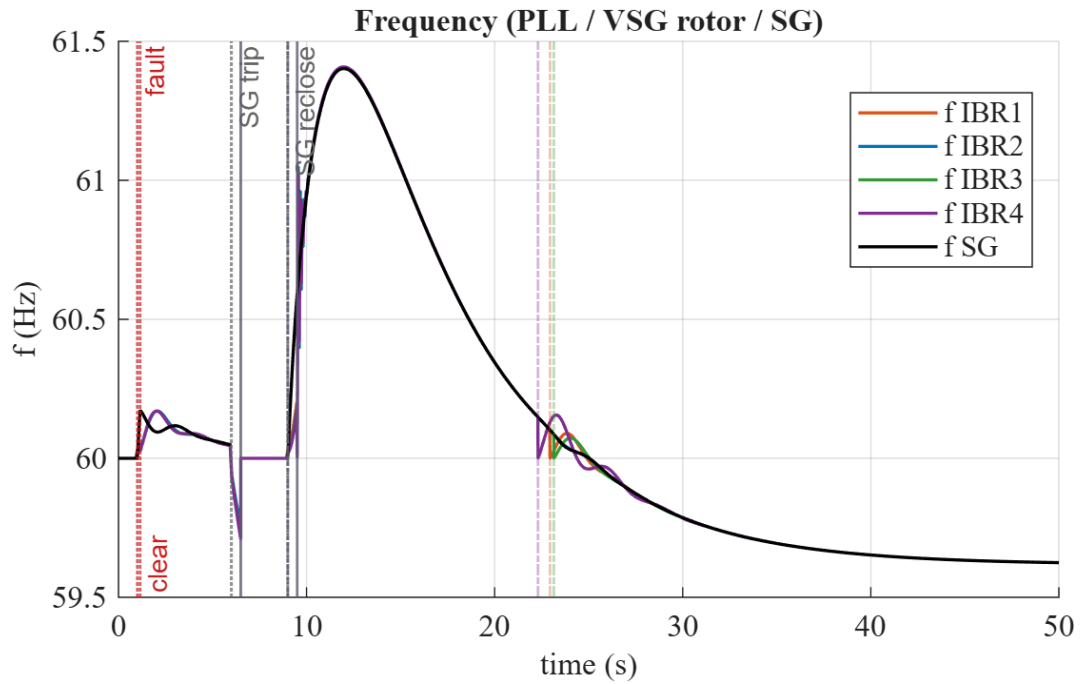


Figure 5: IEEE 14-bus: SG and IBR (PLL / VSG) frequencies; the post-trip excursion peaks near 61.4 Hz and then settles.

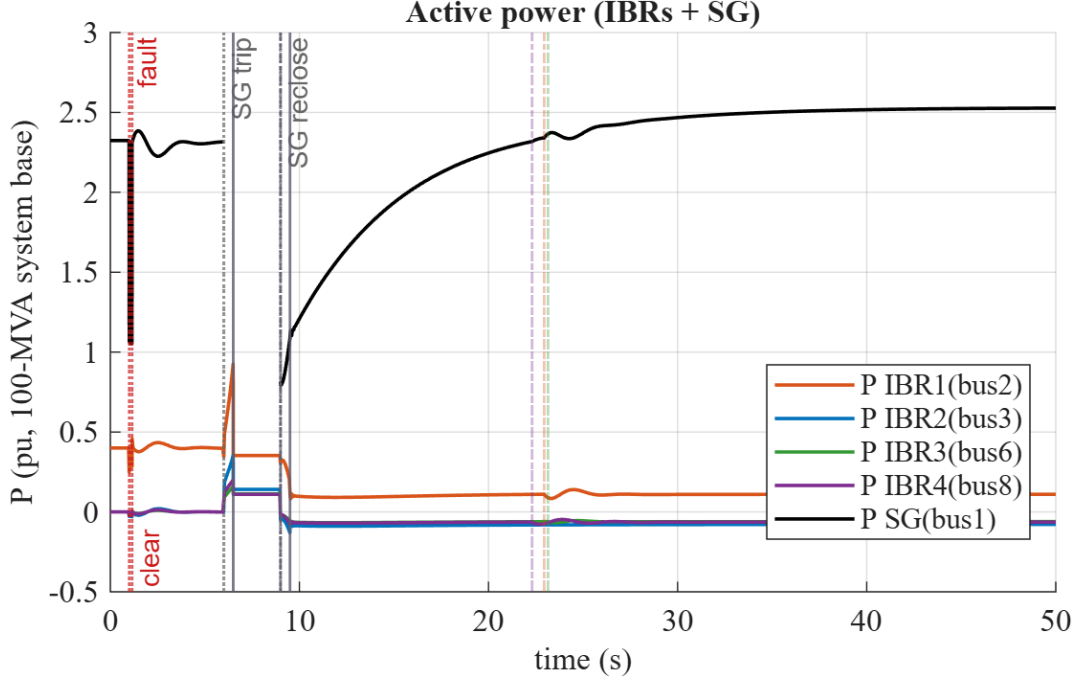


Figure 6: IEEE 14-bus: active-power injections. Pre-event the SG holds ≈ 2.32 pu and IBR1 (bus 2) holds ≈ 0.40 pu; after reclose the SG re-takes ≈ 2.5 pu.

7 Fault switching-outcome map

The switching map is a fault-only sweep (*no* SG trip, $T = 4$ s): a temporary three-phase shunt fault during 1.0–1.15 s is swept over fault reactance and fault bus,

$$X_f \in \{0.02, 0.05, 0.10, 0.20, 0.30, 0.50\} \text{ pu}, \quad \text{fault bus} \in \{3, 4, 5, 9\},$$

and each run is classified into one of four outcome regions (Fig. 7, Table 2): *ride-through* (stayed GFL, no switch), *index-switch* (AGSI++ crosses Γ_{on} so ≥ 1 IBR forms GFM then reverts), *diverge / no post-clear recovery* (the run diverged, or the minimum voltage *measured after the fault is cleared* stays below the 0.25 pu collapse floor), and *non-converged* (fail-closed: the corrector did not converge, so no physical outcome is claimed). The collapse test deliberately uses the **post-clear** minimum voltage: while a low-impedance shunt fault is applied the faulted bus is pulled down *by construction*, so the during-fault sag carries no information about survival. Over the 24 fault points the tally is **ride-through** = 2, **index-switch** = 22, **diverge** = 0, **non-converged** = 0: every swept fault is cleared successfully and the network recovers to 0.98–1.01 pu, with the AGSI++ supervisor forming GFM on all but the two mildest points.

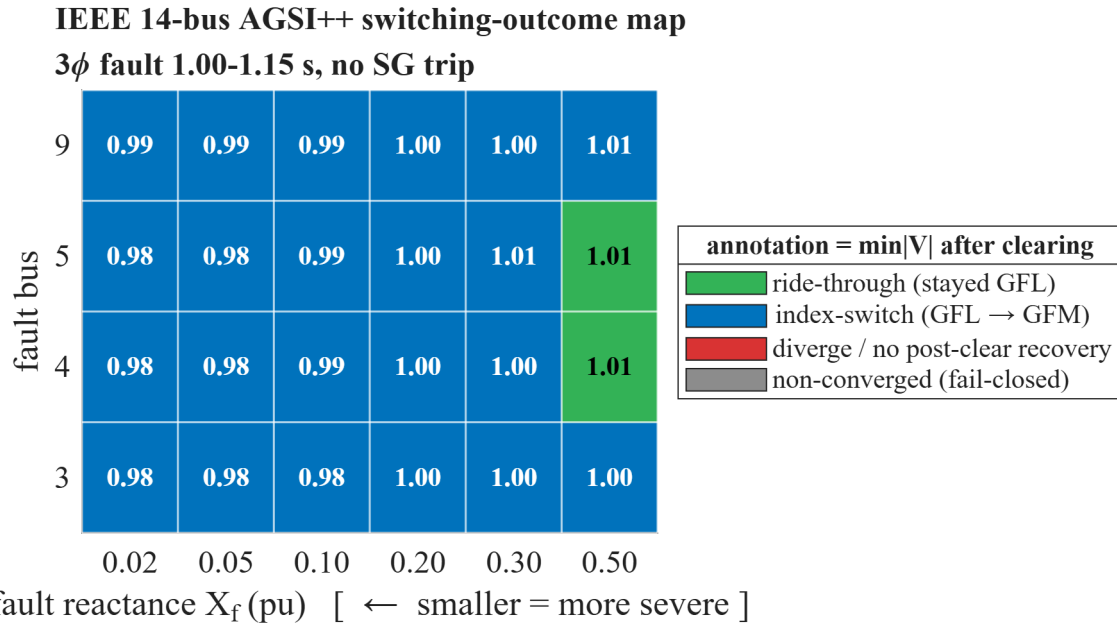


Figure 7: IEEE 14-bus AGSI++ switching-outcome map: fault reactance X_f (x, smaller = more severe) versus fault bus (y); colour = outcome class, annotation = minimum $|V|$ (pu) *after* the fault is cleared (the criterion used by the collapse test).

Table 2: Fault switching-outcome sweep (top, 24 points) and load-step sweep (bottom, permanent step at bus 4). Generated by `ieee14_switching_map.m`.

X_f (pu)	Fault bus	Outcome	min $ V $ (pu) (incl. fault)	min $ V _{\text{post}}$ (pu) (after clearing)	Total switches	Newton conv.
0.02	3	index-switch	0.102	0.980	8	yes
0.05	3	index-switch	0.224	0.985	8	yes
0.10	3	index-switch	0.371	0.983	10	yes
0.20	3	index-switch	0.549	1.001	4	yes
0.30	3	index-switch	0.649	1.002	6	yes
0.50	3	index-switch	0.759	1.004	4	yes
0.02	4	index-switch	0.136	0.979	8	yes
0.05	4	index-switch	0.287	0.978	9	yes
0.10	4	index-switch	0.450	0.987	10	yes
0.20	4	index-switch	0.628	0.995	7	yes
0.30	4	index-switch	0.723	0.999	10	yes
0.50	4	ride-through	0.819	1.009	0	yes
0.02	5	index-switch	0.143	0.979	8	yes
0.05	5	index-switch	0.298	0.982	8	yes
0.10	5	index-switch	0.463	0.987	8	yes
0.20	5	index-switch	0.639	0.995	7	yes
0.30	5	index-switch	0.734	1.006	4	yes
0.50	5	ride-through	0.827	1.009	0	yes
0.02	9	index-switch	0.075	0.990	10	yes
0.05	9	index-switch	0.169	0.988	10	yes
0.10	9	index-switch	0.293	0.992	8	yes
0.20	9	index-switch	0.462	0.998	8	yes
0.30	9	index-switch	0.571	1.004	4	yes
0.50	9	index-switch	0.701	1.007	2	yes
Load step	Step bus	Outcome	min $ V $ (pu)	min $ V _{\text{post}}$ (pu)	Total switches	Newton conv.
+10%	4	ride-through	1.008	1.008	0	yes
+30%	4	ride-through	1.003	1.003	0	yes
+50%	4	ride-through	0.999	0.999	0	yes

Physics of the map. Once the collapse test is applied where it belongs — after clearing — a clean *switching* gradient emerges rather than a collapse region. Every one of the 24 faults is survived: the deepest during-fault sags reach $\min |V| = 0.075\text{--}0.14$ pu (bolted-ish $X_f = 0.02$), yet after the breaker clears at 1.15 s the network recovers to 0.98–0.99 pu, so no point is a voltage collapse and every Newton step converged. What the severity actually decides is *whether the*

AGSI++ supervisor commands forming: from $X_f = 0.02$ up to 0.30 at every bus (and up to 0.50 at IBR bus 3 and the electrically weak bus 9) the sag pushes AGSI++ over Γ_{on} , so at least one IBR forms GFM and then reverts $\Rightarrow$ *index-switch* (22 points, 2–10 transitions each). Only the two mildest cases ($X_f = 0.50$ at the stiffer load buses 4 and 5) stay GFL with zero switches $\Rightarrow$ *ride-through*. Fault location matters as expected: at the IBR bus 3 and the weak bus 9 even the mildest X_f triggers switching, whereas the stiffer load buses 4/5 do not. The load-step sub-table confirms the interpretation: permanent +10/30/50% load steps at bus 4 (no SG trip) all ride through with $\min|V| \geq 0.999$ pu and zero switches — a load step is far milder than a fault and never drives an IBR to form.

8 AGSI++ weight-tuning provenance

The hand-set AGSI++ weights and thresholds were checked against an in-house, offline, design-time search (scripts/reporting/optimize_agsi_weights.m; project-owned base MATLAB, **no external optimization solver**, not on any production runtime path). Guided by the EECON49-P4 Bayesian-Optimization *concept*, the search is a uniform-simplex random sampling of the six weights (each ≥ 0 , sum = 1) plus Γ_{on}/Γ_{off} , followed by a coordinate local refine, minimising, for the IEEE 14-bus SG-trip@1 s + reclose@4 s run over $T = 6$ s:

$$J = 12 \cdot \overline{|1 - V_{\min}(t)|} + 0.6 \cdot (\text{total mode switches}) + 0.05 \cdot \max \text{AGSI} \quad (+ 10^5 \text{ penalty on divergence}). \quad (5)$$

The search improves the objective by only 0.4% ($J : 11.959 \rightarrow 11.906$), i.e. the hand-set weights and thresholds are *already near-optimal* for this stable operating point. The optimum leans toward the grid-strength term (w_{SCR}) and drops w_{lock} , consistent with weak-grid GFL instability being the dominant stress, but the gain is negligible and the hand-set values are more physically interpretable and balanced. **Decision:** keep the frozen defaults $[w_V, w_f, w_R, w_P, w_{SCR}, w_{lock}] = [0.25, 0.25, 0.15, 0.10, 0.15, 0.10]$, $\Gamma_{on} = 0.65$, $\Gamma_{off} = 0.35$; the search is retained only as a reproducible tool that *validates* the choice (ASSUMED_DIAGNOSTIC; frozen before use).

9 Summary

On the meshed IEEE 14-bus system, one manual (no-AVR) synchronous generator and four GFL IBRs form a small-signal-stable operating point (28 states, zero RHP eigenvalues, least-damped 0.345 Hz at $\zeta = +0.267$) that stays stable across the whole inertia and coupling sweep; the AGSI++ supervisor rides through a temporary fault as GFL, forms GFM on the SG trip to hold the island, and hands the reference back on reclose while staying bounded and chatter-free. The switching-outcome map shows that all 24 swept faults are cleared and recovered (no collapse point), with fault severity and location deciding only whether AGSI++ commands forming (22 index-switch, 2 ride-through), and the offline weight search confirms the hand-set AGSI++ weights are near-optimal and frozen. The stark contrast with the small-signal-unstable Padiyar two-area case ($\max \text{Re} = +0.058 \text{ s}^{-1}$) shows the difference is topological.

References

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- [2] H. Ding *et al.*, “A novel design for switchable grid-following and grid-forming control,” *IEEE Trans. Sustain. Energy*, vol. 16, no. 2, pp. 1301–1314, 2025.
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- [5] ENTSO-E, “Rate of Change of Frequency (RoCoF) withstand capability,” Network Code RfG implementation guidance document, 2018.